Quantum Money: Uncloneable Currency Using Quantum Mechanics

Wiesner’s Foundational Scheme

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Agenda

- What is quantum money?
- Wiesner’s scheme
- Security definition
- Key attacks
- Bomb-tester attack
- Proof-of-concept simulation
- Limitations and future directions

What Is Quantum Money?

- Classical money can be duplicated
- Quantum money uses qubits for uncloneability
- Goal: Prevent double-spending via physics

Wiesner's Quantum Money Scheme

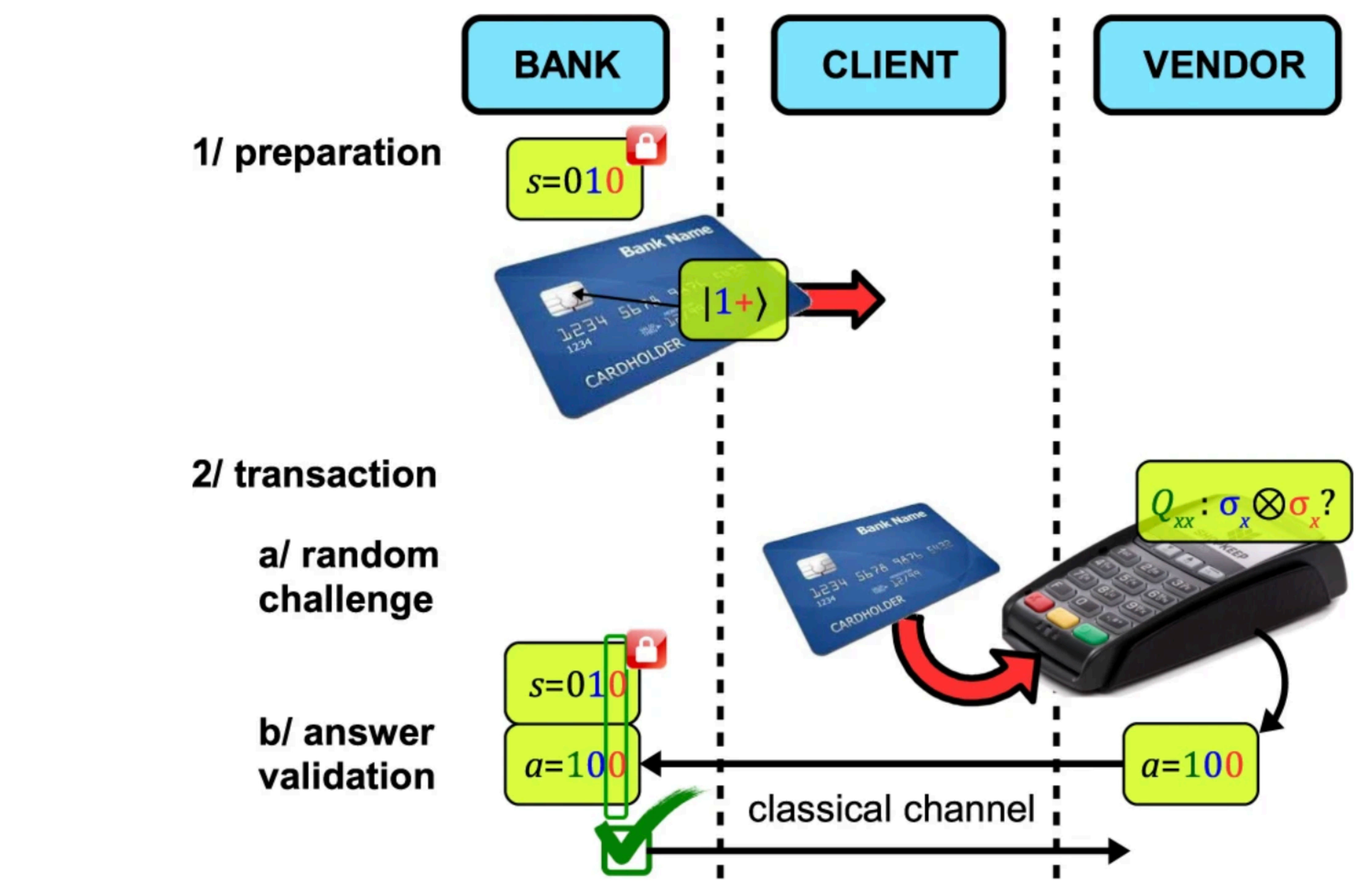
- Bank secretly chooses bases for each qubit
- Four possible states per qubit (BB84 states)
- Serial number is public; preparation key is private

$$
\begin{aligned}
|\psi_{00}\rangle &= |0\rangle, \\
|\psi_{10}\rangle &= |1\rangle, \\
|\psi_{01}\rangle &= |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \\
|\psi_{11}\rangle &= |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle.
\end{aligned}
$$

Verification Process

- Bank uses secret key to measure in correct bases
- Accept if measurement matches original preparation
- Verification is destructive in basic model

From: Experimental investigation of practical unforgeable quantum money



The No-Cloning Principle

- Impossible to copy arbitrary unknown quantum state
- Copying attempt disturbs the state
- Bank detects errors during verification

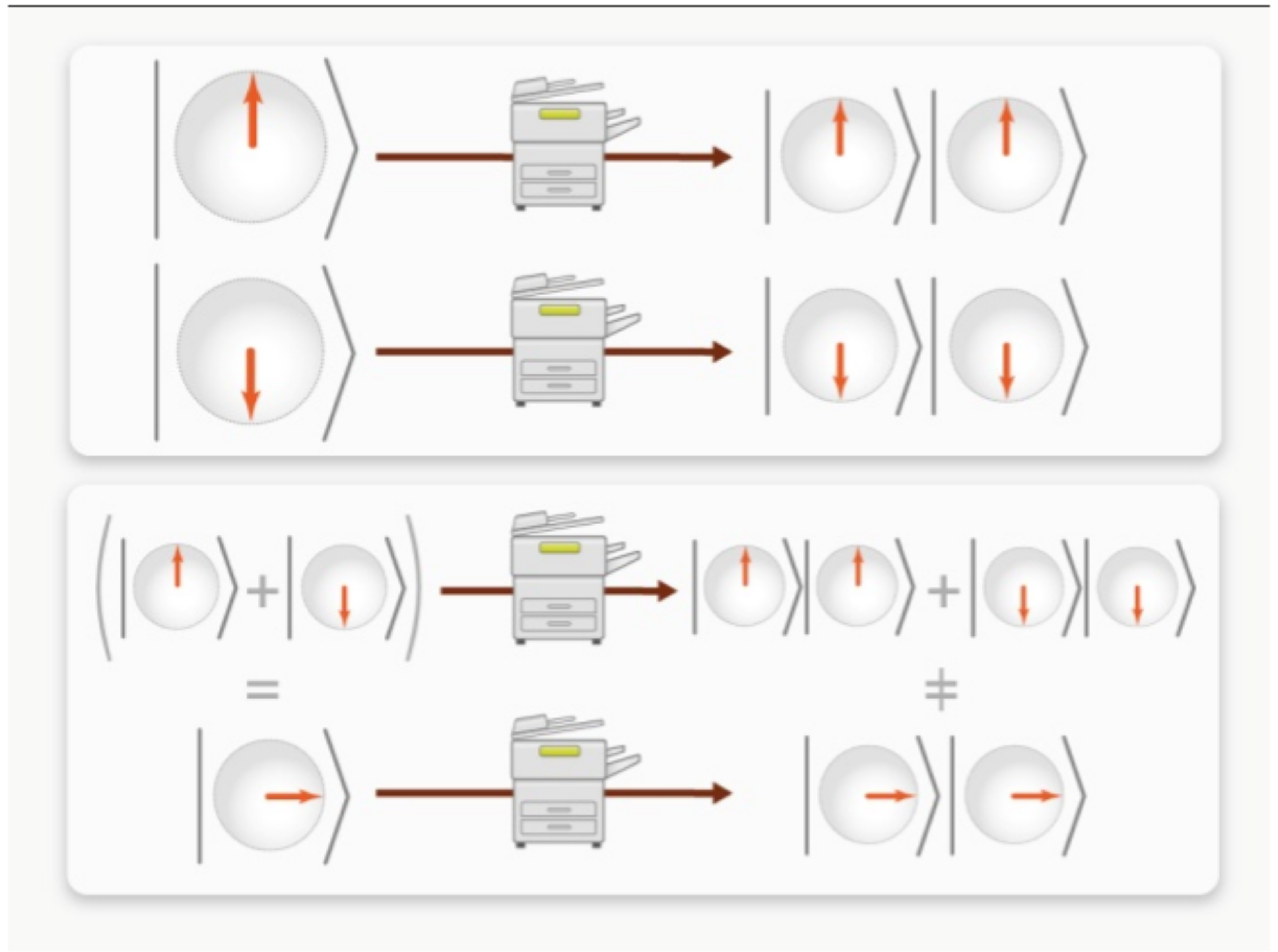


FIG. 17. Illustration of the No-cloning theorem. Any copy machine capable of successfully cloning states $|0\rangle$ and $|1\rangle$ would provide an incorrect result for the superposition state $|0\rangle+|1\rangle$. Hence no perfect quantum cloning machine can exist.

Security Definition – The CLONE Game

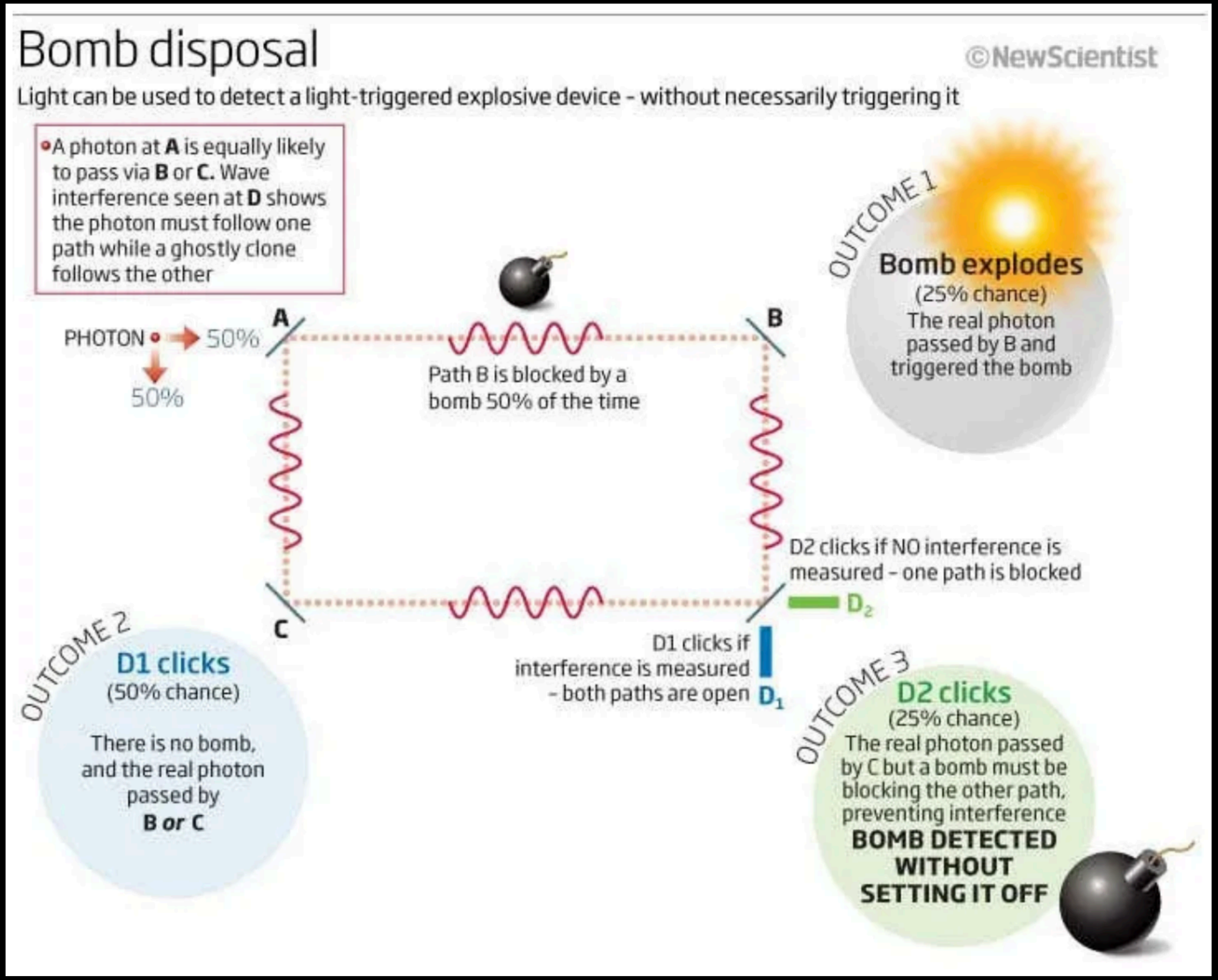
- Adversary receives one valid bill
- Must return two valid copies
- ε-secure if success probability $\leq \varepsilon$

Attacks and Security Scaling

- Best single-qubit attack succeeds with probability 3/4
- n-qubit bills reduce success to $(3/4)^n$
- Exponentially small for $n = 1024$

The Bomb-Tester Attack

- Exploits returning valid bills after verification
- Uses tiny rotations and repeated checks
- Quantum Zeno effect allows learning secret bases



Proof-of-Concept: QuNetSim Simulation of Wiesner's Scheme

- Exact implementation of Wiesner's protocol (Chapter 3)
- Bank prepares & distributes quantum money
- Eavesdropper (Eve) performs Man-in-the-Middle attack
- Bank verification detects tampering → "MONEY IS INVALID"

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Starting transfer
Customer is awaiting serial number and qubits that represent the money
Banker is preparing and distributing qubits
Banker is verifying the money from customer
Banker waiting for serial
Customer received money
Customer is getting his money verified
Customer is verifying the money
Serial received by Bank
Eavesdropper applied X to qubit sent from Customer to Bank
Eavesdropper applied X to qubit sent from Customer to Bank
Eavesdropper applied X to qubit sent from Customer to Bank
Eavesdropper applied X to qubit sent from Customer to Bank
Eavesdropper applied I to qubit sent from Customer to Bank
Eavesdropper applied X to qubit sent from Customer to Bank
Eavesdropper applied I to qubit sent from Customer to Bank
Eavesdropper applied I to qubit sent from Customer to Bank
MONEY IS INVALID
```

Limitations and Public-Key Quantum Money

- Private verification only (bank must participate)
- Public-key quantum money remains open
- Noise and implementation challenges

Historical Context and Conclusion

- First idea in quantum cryptography (1970s)
- Inspired quantum key distribution
- Physics-based security is powerful yet challenging

References

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Questions?